

The Truth Axis is a Semantically Signed Mixture: Per-Category Geometry and a Contract for Sparse Decomposition

Working note — foundations for the sparse chapter, with registered cross-family
falsification pending

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Abstract

The companion paper measured that the one-dimensionality of the truth axis is knowledge-dependent; the anatomy notes measured who writes it and how. This note asks what the axis is *made of*. Grouping CounterFact by Wikidata relation and fitting per-category truth axes at the peak block, we find, on two models (Qwen2.5-1.5B/3B): (1) the axes form a **semantically signed geometry** that replicates across scale nearly cell-for-cell — language-family relations (mother tongue, languages spoken, citizenship) cohere positively (signed cosines $+0.22$ to $+0.38$), while geography (continent) is *anti*-aligned with that block (-0.09 to -0.20 ; confirmed independently by below-chance cross-category transfer, down to AUC 0.26) and sport-position is near-orthogonal; (2) on the 3B, the citizenship axis **switches sides between depth levels** — aligned with geography at the embedding level, with the language block at the peak — so the peak structure contradicts rather than inherits the lexical surface; (3) the FFN’s class-signed write directions decode the category at 69–73% (chance 20%, permutation $p = 0.0099$) against a lexical baseline of $\sim 52\%$: the network *writes* truth along category-dependent directions, twenty points beyond what target-token identity explains. The global truth axis is therefore a mixture with measurable droplets. We deliver the theorist’s handoff: an exported *measured dictionary* (per-category oriented axes, global axis, FFN write centroids, $K \times d$ parameters) and a four-criterion registered contract that any trained sparse autoencoder must satisfy — including beating the measured dictionary — before its parameters are justified. Evidential scope is declared plainly: one model family so far; the cross-family falsification predictions for Llama-3.2 are registered in Section 8 before any such run.

1 Scope and honesty statement

Everything here is measured on **two models of one family** (Qwen2.5-1.5B and 3B) and one dataset (CounterFact, pinned revision), with the shields of the series: held-out cross-validation over pairs, permutation nulls, float32 decomposition gated by additive identity checks (median relative error $1.14/1.22 \times 10^{-7}$), pre-registered predictions, and no automatic verdicts. The signed-geometry magnitudes are small ($|\cos| \leq 0.38$) and two of five categories have weak within-axes (AUC 0.54–0.59), which attenuates their cosines; error bars via bootstrap over pairs are future work. A determinism defect in the decoding fold assignment (salted string hash) was found by cross-run comparison and fixed (process-stable CRC32); canonical numbers are from the fixed tool. Whether this geometry is a property of truth in LLMs or of the Qwen recipe is *not yet known*: Section 8 registers the discriminating predictions, so that the pending Llama measurement is a falsification test rather than additional data.

2 Design

Top- $K=5$ CounterFact relations by unique pairs, 60 pairs each (600 sentences per model): P103 mother tongue, P1412 languages spoken, P27 citizenship, P30 continent, P413 sport position. Per-category truth axes are fit at the peak block (15/16) exactly as the global axis of the companion paper, oriented true = positive *within their own category*, so that the **sign** of a cross-category cosine is information: negative means the shared direction reads the other category’s truth backwards. Three measurements: (P1) signed cosine matrices at the peak and, as a surface control, at an early block (2); (P2) the held-out transfer matrix (row axis $\rightarrow$ column data; diagonal = within-category CV); (P3) nearest-centroid decoding of the category from the FFN’s class-signed write direction Δf at peak+1 (unit-normalized per pair; joint held-out folds; 100-permutation null), against the same decoder on early-block residual differences (the lexical baseline: targets and templates differ by relation, and that surface must not be credited to truth geometry).

3 The signed geometry, and its cross-scale replication

The truth axis is a semantically signed mixture: per-relation truth axes (signed cosines) and category decoding of FFN writes

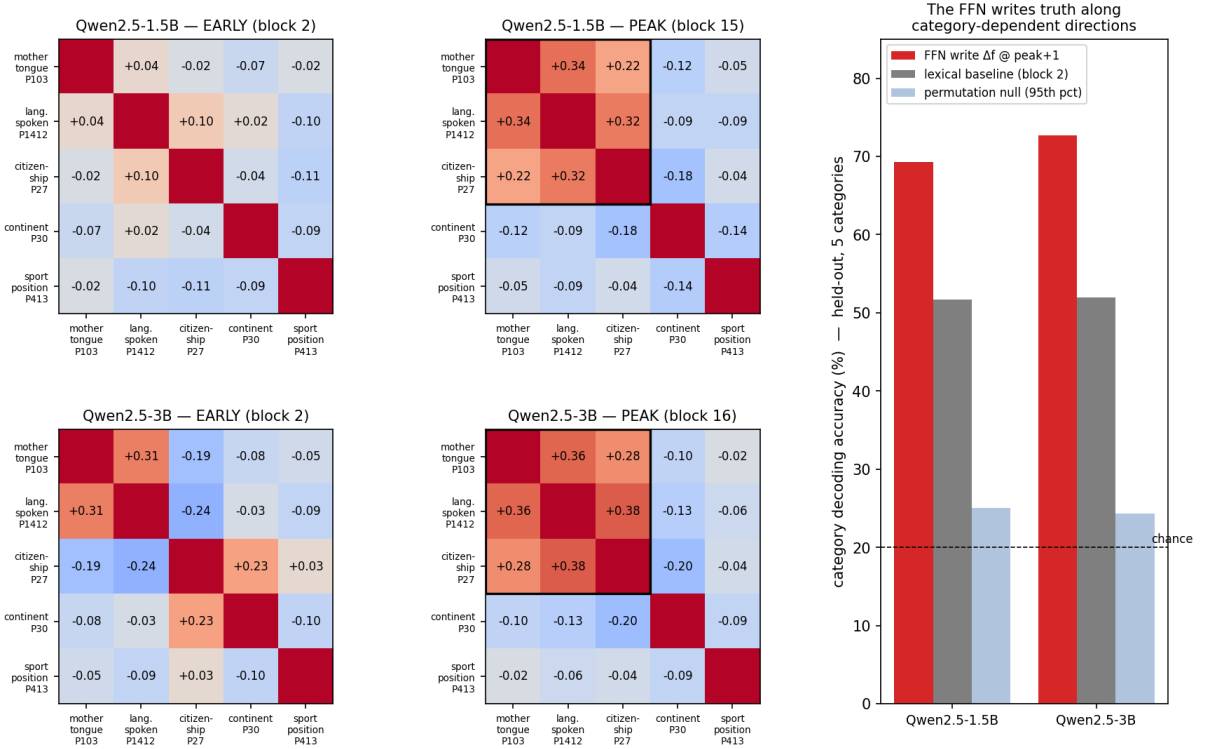


Figure 1: Left 2 $\times$ 2: signed cosines between per-category truth axes, early block vs. peak, both models (language block outlined at the peak). Right: category decoding of the FFN’s write directions vs. the lexical baseline and the permutation null.

At the peak, the language-family block coheres (+0.22 to +0.38), continent is anti-aligned with it (−0.09 to −0.20), and sport-position is near-orthogonal to everything. The same signed pattern appears on both models nearly cell-for-cell (e.g. P27–P30: −0.18 vs. −0.20; P1412–P27: +0.32 vs. +0.38): noise does not replicate signs across models. The anti-alignment is confirmed by an independent statistic: cross-category transfer into and out of P30 falls *below chance* (AUC 0.24–0.46), i.e. a language-truth axis reads geography-truth backwards. Within-category transfer

exceeds cross on both models (0.665 vs. 0.483; 0.687 vs. 0.520), with the cross mean hiding the structure: high inside the semantic block (up to 0.84), at or below chance outside it.

The citizenship reorganization. The surface control is clean on the 1.5B (early cosines flat, $|\cos| \leq 0.11$: the structure emerges with depth). On the 3B it is better than clean — it is *contradictory*: at the embedding level P27 (citizenship) sides with geography (+0.23 with P30, $-0.19/-0.24$ against the language pair, consistent with country tokens resembling place tokens), and by the peak it has switched sides (+0.28/+0.38 with the language pair, -0.20 against P30). The model re-files citizenship from a fact-about-places to a fact-about-persons as processing deepens. A peak structure that contradicts the early one cannot be inherited lexical surface.

4 The FFN writes truth along category-dependent directions

Decoding the relation from the unit write direction Δf at peak+1: 69.3% (1.5B) and 72.7% (3B) against chance 20%, permutation null 95th percentile $\approx 25\%$, $p = 0.0099$ on both. The lexical baseline (same decoder on early residual differences) reaches 51.7%/52.0% — itself above chance, as it must be (targets differ by relation) — leaving ~ 20 points of category structure in the *write directions* beyond token identity, stable across scale. Read together with the anatomy series: the component that overwrites truth frames does so along category-dependent directions. The mixture is not only in the representation; it is in the writing.

5 Interpretation: the mixture, and what it means for sparsity

The companion paper argued from capacity that unknown-fact truth is dispersed in superposition; this note measures the corresponding structure on *known* facts: the global axis is a signed mixture — a shared core within semantic neighborhoods, insularity and partial anti-alignment across distant ones. Three consequences. First, the knowledge-dependent gap gains a candidate anatomy: a single global cut loses exactly the category-specific tails measured here. Second, the anti-alignment is a category-level relative of the polarity structure of Bürger et al.: the orientation of the shared direction with respect to truth is category-dependent. (Whether this reflects deep organization or statement-form differences across relation templates is an open confound, declared here.) Third, sparsity now has a measured target rather than a hope.

6 The theorist’s handoff: a measured dictionary

`dictionary_export.py` extracts and saves, per model, the *measured dictionary*: the K per-category oriented axes, the global axis, and the K FFN write centroids ($[K \times d]$ tensors, $\sim 15k$ parameters at $K=5$, $d=1536$), with the signed matrices and full metadata (model, blocks, revision, seed, decoding statistics) in one loadable bundle. A K -direction readout built from it — project the peak-block state on the axes — is a truth/category monitor whose baseline performance is already quantified above. This is the deliberate division of labor: the theorist ships the metre; the training is engineering.

7 The sparse contract: four registered criteria

A sparse autoencoder trained at the peak is justified only if it satisfies, against matched nulls:

1. **Alignment:** learned features align with the per-category axes above a matched random-direction null;

2. **Geography:** feature sharing respects the semantic structure — shared features within the language block, distinct ones for continent and sport-position;
3. **Signs:** the decomposition reproduces the signed geometry — the shared component loads with inverted sign on P30 — and, the finest test, the citizenship side-switch between early and peak;
4. **Beat the measured dictionary:** on class-signal reconstruction and category decoding, the trained dictionary must outperform the K measured directions of Section 6, or its parameters are unjustified.

Either outcome is content: a dictionary that passes is interpretable because it reproduced independently measured geometry; one that fails is a publishable negative on the SAE hypothesis at this site.

8 Registered cross-family predictions (turning the next run into a falsification)

Written before any Llama measurement of this geometry. On Llama-3.2-3B (peak block 9, write layer 10), `categories.py` with identical settings should yield: (R1) a positive language-family block, signed cosines in $[+0.15, +0.45]$; (R2) P30 anti-aligned with that block (negative cosines and below-chance transfer in both directions); (R3) Δf decoding exceeding the lexical baseline by at least 10 points with permutation $p < 0.05$. The early→peak citizenship switch (R4) is registered as a lead, observed on one model. If R1–R3 hold, the geometry is cross-family and this note is released to the main series; if they fail, the geometry is a family trait and the note is released saying so. The measurement is scheduled; it is a falsification test *because* these lines precede it.

(R5, the arrangement law.) With $K=8$ relations measured identically on Qwen2.5-3B and Llama-3.2-3B, the off-diagonal signed cosines of the two peak matrices correlate positively beyond a category-label permutation null (Mantel-style, $p < 0.05$). If confirmed, the arrangement of the mixture is a property of the knowledge domain, convergent across families (a signed, truth-specific instance of representation convergence); if refuted, the arrangement is recipe-dependent — either outcome is a law-shaped answer. (R6, lead: the citizenship early→peak side-switch of Sec. 3 replicates on Llama — the convergence is functional, not lexical.)

9 Limitations

One family; five relations; 60 pairs each (two weak within-axes attenuate their cosines); one dataset and statement form per relation (the P30 anti-alignment could partly reflect template syntax — crossed-template controls as in the framing study are the known cure); descriptive cosines without bootstrap error bars; the decoding fold-assignment tolerance is $\pm \sim 1.5$ points (measured across shuffles) against effects of +50 over chance. No claim is made that $K=5$ spans the mixture; the dictionary is a floor, not a census.

10 Reproducibility

```
python categories.py
python categories.py --model Qwen/Qwen2.5-3B --peak 16 --write-layer 17

python dictionary_export.py
python dictionary_export.py --model Qwen/Qwen2.5-3B --peak 16 --write-layer 17
```

```
# registered falsification (Sec. 8), when gated access is re-enabled:  
python categories.py --model meta-llama/Llama-3.2-3B --peak 9 --write-layer 10
```

Scripts `categories.py`, `dictionary_export.py` (in `src/`, importing `truth_probe.py` and `anatomy.py`);
figure `categories_geometry.png`. Dataset revision pinned as in the series. License: CC BY 4.0.

References

- [1] F.K. Viciomini. *How Much of Truth Fits on a Single Axis?* Working paper, June 2026. Zenodo DOI 10.5281/zenodo.20938285.
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